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Challenge-aware U-net for breast lesion segmentation in ultrasound images

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ABSTRACT

Deep learning methods can enhance the efficiency of tumor segmentation in breast ultrasound (BUS) images. However, noise interference, small tumors, and blurred boundaries can reduce segmentation accuracy. We design a three-branch challenge-aware U-net (CAU-net) to address these main challenges in BUS images. Our CAU-net extracts the features from three challenge-aware encoders in parallel first. Secondly, we propose an adaptive aggregation layer (AAL) to merge the multi-scale features of three challenging branches, enabling the network to adaptively handle different breast lesion samples with these main challenges. To further enhance the accuracy of segmentation, we introduce the graph reasoning module (GRM) to the network to model the correlation between the channels of the features and acquire the global information in the features. The result of our experiment on two datasets demonstrates the superiority of CAU-net over the advanced medical image segmentation methods. Our code can be downloaded from https://github.com/tzz-ahu.

1. Introduction

Breast cancer poses a significant health risk for women, thus, making early detection is essential. Ultrasound is a widely used, safe, and affordable diagnostic tool for breast examination [1], but the manual diagnosis of lesions on breast ultrasound (BUS) images is time-consuming. Automatic lesion segmentation in BUS images can have an effect on more efficient and accurate diagnosis, attracting increased research attention. Early methods [2–6] mainly rely on manually extracted features or prior knowledge, making it difficult to accurately partition the lesion region in different challenging situations. Since the fully convolution network (FCN) [7] appears, many deep learning techniques have been developed [8–13], achieving commendable results of BUS image segmentation. Then, Yap et al. [14] design comprehensive experiments to systematically evaluate the performances of FCN-based models on BUS images, making this field obtain more attention.

Caused by a variety of factors such as ultrasound imaging noise, the complex structure of human tissue, and various tumor growth, there are some main challenges in tumor segmentation in BUS images, such as noise interference (NI), small tumors (ST), and blurred boundaries (BB). Ning et al. [15] highlight the difficulty of accurately identifying the lesion in BUS images due to noise interference (NI). In addition to noise interference (NI), we find small tumors (ST) and blurred boundaries (BB) also have not been solved well. The samples of NI, ST, and BB challenges are shown in Fig. 1 from left to right, where the contour of the ground truth is plotted with the red line. Therefore, how to solve the above challenges is crucial for improving the effectiveness of computer-aided diagnosis (CAD) on BUS images.

Recently, for various BUS tasks, many methods have noticed these common challenges and tried to handle them. Almajalid et al. [17] utilize contrast enhancement and noise reduction techniques to advance the performance of tumor segmentation in BUS images. These preprocessing techniques can improve the result to a degree, but they lead to incorrect feature representation of the lesions, potentially. Small tumors, which are often early breast lesions, with a low proportion, but they are worthy of attention. To detect the small tumors (ST), Shareef et al. [18] introduce a 'small tumor'-aware network, which contains a dual encoder with row–column-wise kernels to learn a better representation of BUS images. However, the effectiveness of this method relies on the ideal vertical stacking and unwrapping structure of the BUS image. When the distribution of glands is irregular, the performance of this approach will be influenced. For the problem of blurred boundary (BB) of tumors in BUS images, Xue et al. [19]

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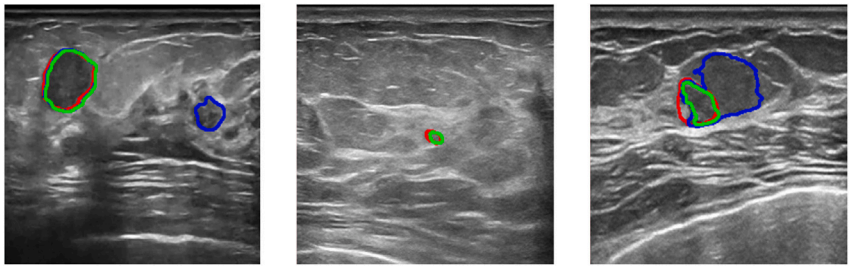

Fig. 1. The segmentation results of U-net [16] and the proposed CAU-net on the three challenge samples correspond to noise interference (NI), small tumors (ST), and blurred boundaries (BB) from left to right. The contours of the ground truth and the results of U-net and CAU-net are plotted with red, blue, and green lines, respectively. When performing tumor segmentation on these challenging samples, U-net generates false segmentation or missed segmentation. However, our method can accurately identify the location of the tumor and produce segmentation results close to the ground truth mask. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

develop a boundary detection module to enhance the boundary details of objects by additionally learning edge features. However, the quality of the learned edge features still depends on the extracted features of the object. Moreover, the aforementioned methods can only handle a single challenge and are unable to adapt to the intricate scenario where multiple challenges exist simultaneously. Therefore, it is important to design a segmentation method that can effectively tackle multiple challenges. Qi et al. [20] introduce an attribute-based network for single target tracking. However, all attribute branches share the same backbone in the shallow layer; thus, obtained features in a specific attribute with only a single scale are not suitable for the encoder–decoder framework. Moreover, the implementation of feature fusion with a simple convolution layer cannot generate an effective and robust representation for challenging samples. Therefore, we redesign the challenge-aware encoder by implementing a challenge branch with an exclusive pre-trained backbone to produce multi-scale features. We also develop an adaptive aggregation layer to learn a better appearance representation of the target lesion with the arbitrary challenge.

Furthermore, to achieve accurate pixel-level prediction in the segmentation task, it is crucial to obtain global contextual information in BUS images. Xue et al. [19] propose a global guidance network, which integrates multiple attention strategies to improve the performance of segmentation. However, the feature fusion strategy in this method generates features with a certain amount of noise, which leads to attention mechanisms that do not yield the best results. Recently, Chen et al. [21] focus on malignant tumor samples in BUS images. They introduce a novel attention module to replace convolutional layers in the encoder and facilitate a more robust representation learning process. Nevertheless, replacing the entire convolution operation with an attention mechanism requires substantial computational resources. These works further inspire us to incorporate global contextual information into the challenging breast tumor segmentation tasks. We anticipate that such integration can improve the overall performance of segmentation.

Based on the above analysis, in this paper, we propose a novel challenge-aware-based network for breast lesion segmentation in BUS images. Unlike the traditional strategy that individually handles different types of challenges, we train specific encoding branches for three different challenges: noise interference (NI), small tumors (ST), and blurred boundaries (BB). Such a design enables more accurate perceptions of various challenges and lessens the influence of tumor diversity under each challenge. Next, facing multi-branch encoders, we naturally design an adaptive aggregation layer (AAL) to aggregate the features of different challenge encoders in each layer and input the fusion feature as a skip connection into one corresponding decoder. In this way, by recovering features layer by layer through the decoders, a predicted mask of tumor segmentation with the same size as the input image can be obtained. The unbalanced Unet architecture of multiple-to-one (multi-branching in the encoding stage and aggregation in the decoding stage) can not only adaptively perceive different challenges at multi-scales, but also effectively cope with complex scenarios where multiple challenges exist simultaneously. Moreover, this design also reduces the network complexity caused by the multi-branch model. Finally, to achieve more refined segmentation, we introduce a graph reasoning module that leverages graph convolution to model the relationships between the channels of features of the deepest layer, capturing better global contextual information. This strategy not only smoothens the deep features of Unet, alleviating the overfitting effect after multi-layer encoding, but also more effectively handles the challenging breast tumor segmentation task. Therefore, the multi-branch challenge-aware encoders, the adaptive aggregation, and the graph reasoning are closely related and constitute an inseparable and functionally synergistic whole, which can improve the performance of breast tumor segmentation. The main contributions of this paper include:

- We propose a novel challenge-aware architecture for breast lesion segmentation with multi-branch encoders and adaptive aggregation layers that can utilize the limited samples to adaptively tackle the three main challenges and more complex scenarios.
- We introduce a graph reasoning module that employs the graph convolutional network to model the channel-wise relationships of high-level features, capturing global contextual information to smooth deep-layer encodings and enabling more effective representation of challenging breast tumor images.
- Experimental results on two BUS datasets demonstrate that our network is superior to many state-of-the-art medical image segmentation methods while maintaining suitable computational efficiency.

In the rest of this paper, we introduce some related works in Section 2. Then we explain the details of our network and the loss function in Section 3. Section 4 gives the experimental settings and experimental results. In Section 5, we discuss some failure cases of segmentation in our experiments, and state the future work. Finally, the conclusion is presented in Section 6.

2. Related works

2.1. Tumor segmentation in BUS images

Tumor segmentation in BUS images has received considerable attention. Previous studies have achieved impressive outcomes. For instance, Almajalid et al. [17] optimize the performance of segmentation by employing contrast enhancement and noise removal strategies, but at the expense of destroying the semantic structure of BUS images. Xue et al. [19] design a global guidance module to learn the global context information of multi-layer features through spatial and channel patterns and enhance the segmentation ability by detecting the boundaries of breast lesions. Ning et al. [15] propose a coarse-to-fine network that leverages hyper-pixel images to enhance the structural information

of the original image for segmenting breast lesions. To overcome the challenge of small objects in breast lesions, Shareef et al. [18] introduce a small tumor sensing network for improving the ability of the network to locate small tumors in the BUS image. Recently, research efforts have also focused on malignant tumor samples in BUS images. Chen et al. [21] develop an adaptive attention network to improve the recognition ability of malignant tumors by replacing the original convolution operation with the designed mixed attention network. However, previous studies do not consider the various challenges comprehensively. In this work, based on challenging samples, we propose a challenge-aware network that can adapt to three main challenges.

2.2. Graph reasoning methods

The CNN model is known for its ability to process local information, but for generating dense pixel-level predictions in segmentation tasks, global information is needed as well. Therefore, some researchers have combined CNN and GCN [22] to infer the long-distance relationship in features, improving the segmentation performance. For example, Chen et al. [23] design a global reasoning unit based on a graph, where they map the original spatial features into an interactive space and conduct graph reasoning in the two dimensions that are nodes and channels to learn the global information of features. Lu et al. [24] develop a method to transform the image segmentation problem into a graph node classification problem by treating the image as the graph structure data. Li et al. [25] create a lightweight method that is capable of performing GCN on the CNN feature by developing a data-dependent Laplace function. Meng et al. [26] employ boundary and region information for graph reasoning, manipulating long-range semantic reasoning of different regions through spatial enhancement along object boundaries to learn graph representation. In this work, we design a graph reasoning module that can effectively model the channel relationships of deep-level features and learn the long-distance information of features.

2.3. Segment Anything Model

The Segment Anything Model (SAM) [27] is an advanced deep-learning architecture for general image segmentation, which has received wide attention. However, despite people making efforts to adapt SAM to medical imaging applications, two key challenges remain: first, the SAM struggles with generalization because of a lack of appropriate adaptation strategies; Second, the segmentation process of SAM is semi-automated and relies heavily on precise manual cues. To address these challenges, researchers have recently proposed the SAMUS for image segmentation. Compared with SAM, it has a stronger ability to preserve subtle local information. With the innovative encoder–decoder framework of the large model and cross-branch focus mechanism, SAMUS [27] has made important advances in the field of medical image segmentation. In addition, AutoSAMUS integrates an Auto Prompt Generator (APG) [28], which replaces the manual cue encoder used in previous methods and automatically prompts for embedding generation. Therefore, it reduces the need for manual input and simplifies the segmentation workflow. SAMUS has shown great potential in BUS image segmentation. However, it is worth noting that large models are extremely computing-resource-consuming. In contrast, some classical methods have relatively comprehensive performances [16,29], with less calculation cost.

3. Methodology

In this section, we introduce the proposed challenge-aware U-net (CAU-net) architecture in detail, including the challenge-aware encoder, graph reasoning module (GRM), training strategy, and loss function. Fig. 2 displays the overall framework of CAU-net for breast lesion segmentation. Our CAU-net employs the encoder–decoder structure and the skip connection strategy similar to U-net [16]. Specifically, for the aforementioned challenges in BUS images, we design a novel encoder consisting of three challenge branches. Furthermore, the encoder features used for skip connections at each layer are aggregated by an adaptive aggregation layer (AAL), which enables the network to adapt to different samples with challenges. To further improve the segmentation performance, GRM is subsequently adopted to capture global contextual information within deep-layer features. Finally, the features with rich semantic information are recovered layer by layer through the decoder, resulting in the predicted mask with the same size as the input image.

3.1. Challenge-aware encoder

Tumor segmentation in BUS images confronts three main challenges: noise interference (NI), small tumor (ST), and blurred boundary (BB). These challenges affect the performance of the segmentation of networks to varying degrees, as shown in Fig. 1. To learn a better appearance representation of these various challenging samples, we design a three-branch challenge-aware backbone. Each branch has a pre-trained 'preference' parameter corresponding to the challenging sample. During testing, the network is unaware of the challenge type in the test samples in advance. Simultaneously, a single sample could have multiple challenges. Extracting features from three branches in parallel may degrade the quality of the output compared with a traditional single-branch feature extractor. Therefore, an adaptive aggregation layer (AAL) is designed to appropriately fuse the extracted features from the three challenge branches. Then, the features of each layer extracted by the backbone network are balanced among the three challenge 'preferences', which facilitates the adaptive response of the network to different samples with arbitrary challenges. We utilize ResNet-34 as the backbone for each challenge branch in the network and extract the highest four-level features for adaptive aggregation.

The structure of the adaptive aggregation layer is presented in the upper right corner of Fig. 2. X^l_{NI}, X^l_{ST}, and X^l_{BB} represent the feature of the NI challenge branch, the ST challenge branch, and the BB challenge branch, respectively, extracted by the encoder of layer l. The features of the three challenging branches first go through a 1×1 convolution layer, respectively. Then we use a nonlinear activation layer to learn nonlinear mapping $\widetilde{X}^l_{NI}$, $\widetilde{X}^l_{ST}$, and $\widetilde{X}^l_{BB}$. For the massive interaction between the NI sample and ST sample, we calculate the mutual enhancement feature of $\widetilde{X}^l_{NI}$ and $\widetilde{X}^l_{ST}$. This operation is implemented with matrix multiplication. Then $\widetilde{X}^l_{BB}$ is aggregated with the enhancement feature above. Finally, a convolution layer with 3×3 kernel size and 1×1 kernel size, a batch normalization layer, and a ReLU linear activation layer are used to obtain the encoder feature of layer l for skip connection. The calculation process of X^l can be summarized by the following formulas:

$$\widetilde{X}^l_{NI} = g\left(X^l_{NI}\right), \tag{1}$$

$$\widetilde{X}^l_{ST} = \theta\left(X^l_{ST}\right), \tag{2}$$

$$\widetilde{X}^l_{BB} = h\left(X^l_{BB}\right), \tag{3}$$

$$X^l = CBR\left(\alpha \widetilde{X}^l_{BB} \cdot Softmax\left(\widetilde{X}^l_{NI} \cdot \widetilde{X}^l_{ST}\right)\right), \tag{4}$$

where $g(\cdot)$, $\theta(\cdot)$ and $h(\cdot)$ represent three nonlinear mapping operations, respectively. CBR represents convolution, batch normalization, and ReLU linear activation operation. $\otimes$ represents element-wise multiplication. α is a learnable scaling parameter. In conclusion, through the encoding operations defined by Eqs. (1) to (4), the proposed framework can not only adaptively perceive different challenges at multi-scales, but also effectively cope with complex scenarios where multiple challenges exist simultaneously.

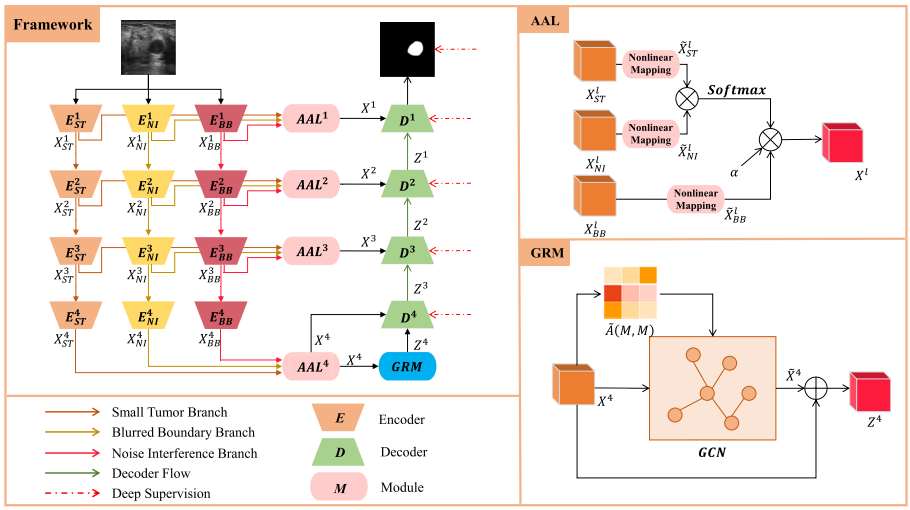

Fig. 2. Overview of the proposed CAU-net architecture.

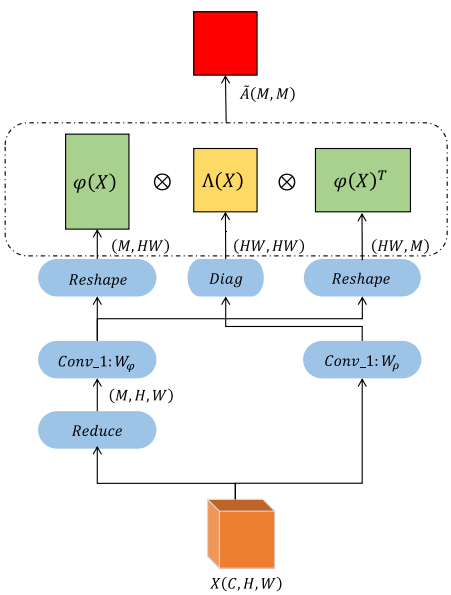

Fig. 3. The calculation process of the similarity matrix $\widetilde{A}$.

3.2. Graph reasoning module

Through multi-layer and multi-branch encoders, our framework effectively perceives different challenges and obtains high-level image representations. However, repeated encoding may also lead to the intensification of noise and overfitting, lacking guidance from contextual information. Recently, graph convolutional networks [22] have attracted widespread attention due to their ability to capture structural information [30–34]. Therefore, in this paper, we introduce a graph reasoning module into the Unet architecture to smooth high-level features extracted from the multi-layer encoders, capturing contextual information and obtaining more representative deep features, thereby further improving BUS image segmentation performance. The detailed reasoning process is described below.

First, a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed based on the features obtained from the challenge perception encoder. Here, each vertex in this graph denotes a channel of features from the last layer AAL, and its vector representation is the channel-specific spatial feature, which is extracted and aggregated by the challenge-aware encoder. Edges $\mathcal{E}$ represent relationships between these channels. Accordingly, the adjacency matrix $\widetilde{A}$ is computed by the self-weighted dot product operation between the corresponding channel-specific spatial features. Fig. 3 illustrates the computation process for $\widetilde{A}$. The specific computing formula of $\widetilde{A}$ is as follows:

$$\widetilde{A} = \varphi(X)\Lambda(X)\varphi(X)^T, \tag{5}$$

where X is the output feature of the challenge-aware encoder. $\varphi(X) \in \mathbb{R}^{M\times HW}$ is a linear embedding which simultaneously reshapes the original feature X and reduces its dimension. M is the reduced channel after transformation. $\Lambda(X) \in \mathbb{R}^{HW\times HW}$ is a diagonal matrix that contains spatial information of the original features. It is defined as $\Lambda(X) = diag(vec(X))$, which $vec(\cdot)$ turns X into a vector through the global pooling and reshaping with a 1×1 convolution. The self-weighted dot product calculation flexibly models the relationships between different channels. This data-driven graph structure enables the model to capture long-range contextual dependencies, facilitating the subsequent reasoning and segmentation.

Second, building on this graph, we construct the Laplacian matrix, using channel-spatial attention to ensure that GCN learns long-range dependencies. Here, the symmetric normalized Laplacian matrix can be defined as follows [25]:

$$\widetilde{L} = I - \widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}, \tag{6}$$

where $\widetilde{D}$ is the degree matrix of $\widetilde{A}$ and $\widetilde{A} \in \mathbb{R}^{n\times n}$. We set $n = M$, which denotes the reduced channel of the input features.

Finally, the graph reasoning process can be formulated as follows:

$$Z = \sigma\left(\widetilde{L}X\Theta + X\right), \tag{7}$$

where Z is the output feature of GRM, X is the output feature of the challenge-aware encoder. Θ is a trainable weight matrix and $\sigma(\cdot)$ is the ReLU activation function. Accordingly, $\widetilde{X} = \widetilde{L}X\Theta$ is the output of GCN using the designed Laplacian matrix $\widetilde{L}$ above. Unlike traditional GCNs, our approach builds a Laplacian matrix to ensure that the GRM captures both spatial and channel-wise relationships, which are crucial for segmenting complex breast lesions. The refined features from GRM are then progressively recovered through the decoder to generate the final segmentation mask at the original resolution.

3.3. Decoder flow

The deep features enhanced by graph reasoning are processed through a four-layer decoder that progressively restores the resolution to produce the final prediction map. Each decoder block contains two

Conv+BN+ReLU (CBR) operations. Between these, features from the previous decoder layer are concatenated with those from the corresponding encoder layer by the skip connection. The nearest neighbor up-sampling with a scale factor of 2 is then applied to scale up the current layer's output for the next layer. Each decoder layer also generates an intermediate prediction map for supervision. Finally, the last decoder layer's output passes through a 1×1 convolution and a 4x up-sampling to produce the final prediction, matching the size of the input BUS image.

3.4. Training algorithm

It is worth noting that the proposed network has two problems in its implementation. First, the loss of the training samples, which can have arbitrary attributes, must be backpropagated through all three challenging branches [35]. Second, during testing, the attributes of each sample are unknown, and the network must dynamically aggregate the results from the three challenging branches. To address these issues, we train the network in two stages.

In the first stage, we divide the entire dataset into three sub-datasets corresponding to the challenges of ST, NI, and BB, respectively. Then we use the ResNet-34 as the backbone to train on the three sub-datasets. The three training procedures are carried out independently, using the same hyperparameters and loss function.

In the second stage, we initialize the encoder parameters learned in stage one and disregard the decoder parameters. We then use the entire dataset to train the AAL, GRM, and decoder. This training follows the same hyperparameters and loss function used in the first stage. With this two-stage training approach, our network is better equipped to handle diverse samples with varied attributes.

3.5. Loss function

Binary cross entropy (BCE) loss [36] and Dice loss are commonly used loss functions for image segmentation tasks. The formulas for BCE loss and Dice loss are as follows:

$$L_{BCE} = \frac{1}{N} \sum_{i} - \left[y_i log\left(p_i\right) + \left(1 - y_i\right) log\left(1 - p_i\right) \right] \tag{8}$$

$$L_{Dice} = 1 - \frac{2|M_G \cap M_P|}{|M_G| + |M_P|} \tag{9}$$

where N is the total pixels of input image, y_i and p_i are the ground truth and predicted value of the $i^t h$ pixel respectively. M_G is the ground truth mask, M_P is the predict mask, and $|\cdot|$ is the number of pixels belonging to object.

However, as a classification loss function, BCE can produce biased results when the number of background pixels outweighs that of foreground pixels, especially for samples with small targets. This shortcoming makes the evaluation function based on Dice Loss more effective, as it emphasizes the intersection over union (IoU) metric, which can more effectively manage sample imbalance and better assess the internal filling of the target. To leverage the advantages of both BCE and Dice Loss, we apply a loss function that combines both methodologies. In addition, we implement the loss function with a deep supervision approach, which means that the prediction maps generated at each layer of the decoder are subject to supervision. The total loss function can be defined as follows:

$$L_{total} = \sum_{l}^{N} \left(L_{BCE} + \lambda \cdot L_{Dice} \right)^l \tag{10}$$

where N is the total number of layers of the network, and $(\cdot)^l$ is the loss of layer l. λ is the coefficient used to balance Dice Loss, which we set to 0.75 in the experiments.

Table 1
The Sample Distributions of the AMUBUS, BUSI, and BrEaST datasets.

	Benign	Malignant	Normal	Total
AMUBUS	2357	285	No	2642
BUSI	437	210	133	780
BrEaST	154	98	4	256

4. Experiments

In this section, we first introduce the BUS datasets, experiment settings, and evaluation metrics. Then, a number of experiments are conducted to verify the superiority and effectiveness of our network. At last, we show some visual results of the different methods.

4.1. BUS dataset

Currently, publicly available BUS image datasets of breast tumors are very limited. Also, due to insufficient sample size and lack of challenges, these datasets are not suitable for practical use. To fully conduct the experiments on the BUS image segmentation and accurately verify the effectiveness of the proposed CAU-net, we first conduct our experiments on our BUS image database, which is AMUBUS. The details of the database are as follows.

The AMUBUS dataset consists of 2642 ultrasound images obtained from 528 patients at the First Affiliated Hospital of Anhui Medical University using the Resona 7 and Toshiba 660a ultrasound systems. Three to five video frames with the greatest variance in shape are from each ultrasound video (there may be multiple videos per case). All images have not undergone any cropping or data cleaning, which includes a large number of challenge samples (NI, ST, and BB). The dataset also contains images of various sizes, the largest being 1072×756 and the smallest being 256×256. To facilitate the arrangement of the results, we scale all the images to 256×256. Each image in AMUBUS is labeled by three radiologists with at least 3 years of experience, and the quality of the annotations is controlled and refined by a senior department director. To facilitate further research, we will make AMUBUS openly available in future work.

In addition, to fully verify the performance of the method, we also introduce BUSI and BrEaST datasets. The classic BUSI database is created by Al-Dhabyani et Al. [37]. BUSI collects 600 female patients between the ages of 25 and 75. It includes 780 images (including 133 normal samples) obtained from Baheya Hospital using the LOGIQ E9 and LOGIQ E9 Agile ultrasound systems. The average size of images in BUSI is 500 x 500. To unify the data format, we adjust the image size in BUSI to 256×256. As for the latest BrEaST dataset [38], it has been used in the cancer image download archive (TCIA), and can also be used in its special web page to see. The data comes from 256 adult female patients aged 18 to 87 years. A total of 197 biopsies are performed (77% of the dataset), and 98 cases of breast cancer are confirmed according to BIRAD criteria. Of these, the biopsy results for BIRADS 3 (12 cases) are diagnoses made prior to follow-up ultrasound scans (images obtained during ultrasound scans). Finally, the dataset includes 98 cancers, 154 benign lesions, and 4 normal tissue images. Table 1 shows the sample distribution of the three datasets. It can be seen that the amount of data of AMUBUS is much larger than that of BUSI and BrEaST. All experiments in this paper are conducted on the above three datasets. Detailed experimental procedures and results will be described in the next section.

4.2. Experiments settings

We evaluate our method on two datasets of BUS image segmentation, conducting comparisons with leading methods and performing ablation studies. We mainly rely on the AMUBUS dataset as a pre-training resource in comparison experiments, considering that AMUBUS has

Table 2
Effect of the components proposed in CAU-net (mean). The best results are highlighted in red, while the secondary ones are marked in blue.

CA	AAL	GRM	Acc (%)↑	Se (%)↑	Dice (%)↑	IoU (%)↑	HD (mm)↓
			98.17	86.34	82.91	73.65	6.42
✓			98.43	87.54	83.89	74.82	6.33
✓	✓		98.60	88.13	84.73	75.69	6.25
		✓	98.19	87.26	83.85	74.28	6.40
✓	✓	✓	99.22	88.61	85.49	76.26	6.13

a larger amount of data and a richer set of challenges. We divide the AMUBUS into three sub-datasets for each of the three challenge branches (ST, NI, and BB), each comprising 637, 745, and 472 images, respectively. The complete network is trained on four-fifths of the total dataset and evaluated on the remaining one-fifth. For the BUSI dataset, we employ the same train–test split ratio, but due to its small quantity, we do not divide it into challenge subsets; Instead, we train using the above-mentioned pre-training of the challenge branches. Ablation studies are conducted only on the AMUBUS to establish the efficacy of the strategies in CAU-net. Finally, we show some visual comparison results.

We deploy our model on one NVIDIA RTX 3090 GPU with a development environment of Python 3.8 and PyTorch 1.7.0. Adam optimizer is utilized to accelerate the training and weight decay is set as 0.0001. For the hyperparameters, the initial learning rate, batch size, and the number of epochs are set as 0.0001, 32, and 150, respectively.

4.3. Evaluation metrics

To verify the performance of CAU-net from multiple aspects, we employ five metrics with different measurement rules. They are Accuracy (Acc), Sensitivity (Se), Intersection over Union (IoU), Dice index(Dice), and Hausdorff distance (HD). The first two are classification indices. Dice and IoU are widely used in the segmentation task. HD is sensitive to the boundary but does not take into account the shape of the target. These metrics are respectively defined as follows:

$$Acc = \frac{|I| - |M_G \cup M_P| + |M_G \cap M_P|}{|I|} \tag{11}$$

$$Se = \frac{|M_G \cap M_P|}{|M_G|} \tag{12}$$

$$Dice = \frac{2|M_G \cap M_P|}{|M_G| + |M_P|} \tag{13}$$

$$IoU = \frac{|M_G \cap M_P|}{|M_G \cup M_P|} \tag{14}$$

where I is the input image, M_G is the ground truth mask, M_P is the predict mask, and $|\cdot|$ is the number of pixels belonging to object.

$$h\left(S\left(M_G\right), S\left(M_P\right)\right) = \max_{s_g \in S(M_G)} \min_{s_p \in S(M_P)} \|s_g - s_p\| \tag{15}$$

$$HD = \max\left(h\left(S\left(M_G\right), S\left(M_P\right)\right), h\left(S\left(M_P\right), S\left(M_G\right)\right)\right) \tag{16}$$

where $S\left(M_G\right)$ and $S\left(M_P\right)$ denote the surface pixel set of M_G and M_P respectively, s_g and s_p are respectively an arbitrary pixel in M_G and M_P, and $\|\cdot\|$ is the shortest Euclidean distance.

4.4. Ablation study

4.4.1. Effect of the Strategies Proposed in CAU-net

In our work, we introduce a challenge-aware backbone (CA) to obtain a better appearance representation of the challenge sample, and an adaptive aggregation layer (AAL) is well-designed to merge features of three branches. Moreover, we develop a graph reasoning module (GRM) to extract global information, enabling the network to achieve better accuracy of segmentation. To prove the effectiveness of the above strategies, we conduct the ablation experiments first.

Table 3
Effect of different challenge branch strategies (mean). The best results are highlighted in red, while the secondary ones are marked in blue.

ST	NI	BB	Acc (%)↑	Se (%)↑	Dice (%)↑	IoU (%)↑	HD (mm)↓
			98.17	86.34	82.91	73.65	6.42
✓			98.40	87.02	83.38	74.13	6.36
	✓		98.18	86.63	83.16	73.69	6.40
		✓	98.17	86.37	83.07	73.70	6.42
✓	✓		98.30	87.15	83.40	74.24	6.35
✓		✓	98.27	87.13	83.23	74.08	6.38
	✓	✓	98.25	86.74	83.10	73.85	6.46
✓	✓	✓	98.43	87.54	83.89	74.82	6.33

The comparison results of our method with various components are summarized in Table 2. The results of the baseline are shown in the first row, while the second row demonstrates the effectiveness of the challenge-aware backbone. As shown in Table 2, the baseline method achieves poor performance. The use of the 'CA' results in significant improvements, supporting the validity of the concept of challenge awareness. While combining 'AAL' with 'CA', the segmentation performance can be further improved, which demonstrates that 'AAL' aids in better aggregating the features identified by each of the three challenge branches. When adding 'GRM' into the baseline only, the network can also get some promotion, emphasizing that the 'GRM' element can learn crucial global information and further enhance the accuracy of segmentation. As expected, our method with all three components achieves the best results. Acc, Se, Dice, and IoU obtained by our method are 1.05%, 2.27%, 2.58%, and 2.61% higher than baseline, respectively, and HD is decreased by 0.29 mm.

4.4.2. Effect of different challenge branch strategies

Our proposed challenge-aware backbone is composed of three challenge branches corresponding to ST, NI, and BB. To explore the different strategies of the combination of the challenge branches, we conduct numerous experiments. The performances of different strategies are reported in Table 3. The results of the baseline are shown in the first row. The second row to the fourth row shows the results of utilizing a single challenge branch, respectively, and the rest of the table demonstrates the results of using multiple challenge branches. For exploring the performance of different challenge branch combinations and excluding the influence of other modules, we do not connect AAL and GRM in the network, and the features of different branches are concatenated when implementing the network with multiple challenge branches.

From Table 3, we can observe different levels of improvement in the performance of segmentation when using the challenge-aware strategy. Moreover, implementing the network by combining ST and NI achieves better performance than ST + BB and NI + BB. We analyze that this result occurs because there are more samples with challenges, both ST and NI, in the dataset than in other cases, which motivates us to compute mutual information to draw connections between challenges when designing AAL. It is easy to see that the complete challenge-aware backbone with three branches achieves the best performance, scoring higher Acc, Se, Dice, and IoU as well as lower HD.

Table 4
Quantitative results (mean ± std) on AMUBUS. The best results are highlighted in red, while the secondary ones are marked in blue.

Method	Acc (%)↑	Se (%)↑	Dice (%)↑	IoU (%)↑	HD (mm)↓
U-net [16]	97.67 ± 0.53	86.18 ± 2.23	80.38 ± 2.47	70.80 ± 2.19	6.68 ± 0.55
SegNet [29]	98.93 ± 0.48	87.34 ± 2.66	82.90 ± 2.12	72.71 ± 1.73	6.73 ± 0.48
DeepLabV3+ [39]	96.52 ± 0.82	84.51 ± 3.57	79.25 ± 3.46	68.90 ± 2.88	7.17 ± 0.76
Unet++ [40]	98.19 ± 0.45	87.60 ± 2.85	83.56 ± 1.79	73.36 ± 1.34	6.43 ± 0.54
PraNet [41]	98.90 ± 0.36	86.82 ± 1.73	83.00 ± 1.83	72.83 ± 1.58	6.85 ± 0.58
RFNet [42]	98.14 ± 0.37	86.63 ± 1.67	83.27 ± 1.47	73.09 ± 1.26	6.68 ± 0.52
TRUnet [45]	98.95 ± 0.26	87.65 ± 1.58	83.74 ± 1.25	74.62 ± 1.04	6.38 ± 0.65
PSPNet [43]	98.11 ± 0.52	88.05 ± 2.01	83.57 ± 2.34	74.56 ± 1.98	6.25 ± 0.51
nnUnet [44]	98.37 ± 0.42	88.18 ± 2.17	84.38 ± 1.92	74.84 ± 1.56	6.18 ± 0.44
MedNeXt [46]	98.47 ± 0.31	88.25 ± 1.32	84.42 ± 0.79	75.30 ± 0.68	6.25 ± 0.51
AAU-net [47]	97.83 ± 0.15	85.21 ± 2.04	82.76 ± 0.48	73.11 ± 1.55	7.88 ± 0.93
EMCAD [48]	98.89 ± 0.15	86.52 ± 1.71	83.55 ± 1.01	74.82 ± 0.83	6.87 ± 0.56
MoSID [49]	98.51 ± 0.22	85.47 ± 1.82	82.91 ± 0.96	72.83 ± 0.92	7.21 ± 0.25
CAU-net	99.22 ± 0.12	88.61 ± 1.61	85.49 ± 0.86	76.26 ± 0.72	6.13 ± 0.45

4.5. Comparison with other segmentation networks

4.5.1. Quantitative comparison

We compare the proposed CAU-net with many leading segmentation methods to evaluate its effectiveness, including the U-net [16], SegNet [29], DeepLabV3+ [39], Unet++ [40], PraNet [41], RF-Net [42], PSPNet [43], nnUnet [44], TransResUnet [45], MedNeXt [46], TRUnet [45], AAU-net [47], EMCAD [48], MoSID [49], ATFE-Net [50], HAU-net [51] and Hybrid-CAF [52]. The methods above are all implemented with the encoder–decoder framework and are designed to enhance the performance of segmentation using distinct strategies. DeeplabV3+ [39], U-net [16], and SegNet [29] are the three most widely used segmentation baselines. Unet++ [40] proposes a redesigned skip connection strategy, allowing the network to take into account multi-level features. PraNet [41] utilizes reverse attention to refine the boundary of segmented tumors. RF-Net [42] proposes a novel residual feedback strategy that enables the network to dig more information from the region of confusion. PSPNet [43] proposes a pyramid pooling module to aggregate multi-scale contextual information, enabling the network to effectively capture both global and local features. nnU-net [44] introduces a self-configuring framework that automatically adapts preprocessing, architecture, and training pipelines to best fit the target dataset, optimizing segmentation performance across diverse medical imaging tasks. TRUnet [45] combines Transformer and CNN to promote the model to consider the information of local and global simultaneously. MedNeXt [46] leverages a hierarchical convolutional architecture inspired by neural operators to effectively capture both local and global information for breast lesion segmentation. Apart from these classic methods above, we also comprehensively compare our proposed method with some recent breast cancer analysis methods in latest two years, such as AUU-net [47], EMCAD [48], MoSID [49], ATFE-Net [50], HAU-net [51] and Hybrid-CAF [52], to thoroughly evaluate their performance and effectiveness. To provide a fair comparison, all the networks were re-trained on three datasets.

The performance of the CAU-net and other methods on AMUBUS are reported in Table 4. The statistics of each metric are declared in mean and standard deviation. In this paper, we propose a novel challenge-aware U-net, which can learn a better appearance representation of the challenge samples. We can observe that CAU-net achieves higher Acc, Se, IoU, Dice, and lower HD. Our method scores 99.22%, 88.61%, 85.49%, 76.26%, and 6.13 mm on five metrics, which improved by 0.27%, 0.43%, 1.11%, 1.42%, and 0.05 mm, respectively, compared with the second best score. It can be seen that the gap of most comparison methods is not so large, except for DeeplabV3+ [39]. This could be due to the lack of a skip connection strategy commonly employed in segmentation networks for medical images, which inspires us to continue using this framework in U-net. However, as mentioned earlier, we make a new design under this framework to fully address various challenges and to learn better global contextual information, so our approach achieves better results.

Table 5 displays the segmentation result of different methods on BUSI. We can observe that our method improves the first four metrics by 0.18%, 0.11%, 0.75%, and 0.89% from the second place with 97.48%, 89.83%, 87.27%, and 79.13%. It is worth noting that although the HD metric is not as good as nnU-net, our method still ranks second among all comparison approaches. Moreover, it outperforms nnU-net in the other four metrics. Compared with the results achieved in AMUBUS, our approach leads to performance boosting on BUSI due to the lack of challenge samples in BUSI, preventing us from fully utilizing our proposed challenge-aware strategy. Moreover, we can observe that all methods achieve lower Se, Dice, and IoU on AMUBUS than on BUSI, which also means AMUBUS is more challenging. To conclude, our CAU-net obtains superior results on two datasets.

To further validate the generalization and effectiveness of the proposed framework, we also conduct comparative experiments on the latest BrEaST segmentation dataset, as shown in Table 6. The results in Table 6 indicate that the proposed CAU-net has an improved performance compared to traditional methods on multiple metrics. Specifically, CAU-net achieves an average Dice coefficient of 77.88%, which is 3.09% higher than U-net and 3.53% higher than DeepLabV3+. Additionally, CAU-net achieves a superior performance in HD (8.55 mm), outperforming the latest models (nnUnet (9.15 mm)), indicating its robustness in handling challenging cases with precise boundary predictions. Furthermore, compared to the advanced MedNeXt method, the CAU-net framework exhibits better overall performance in terms of both segmentation accuracy and sensitivity, showing its superiority even over the latest methods.

Notably, the experimental results on the BrEaST dataset exhibit significant fluctuations compared to the AMUBUS and BUSI datasets. The observed fluctuations primarily stem from the small scale of this dataset (total image number in the BrEaST dataset is 256), which is significantly smaller than the other two datasets (total image number in the AMUBUS dataset is 2642 and total image number in the BUSI dataset is 780, as shown in Table 1). According to statistical principles, a small sample size may not accurately reflect the real population distribution of the data, thus potentially exaggerating data variability and consequently making the impact of individual segmentation accuracy on overall results more pronounced. Additionally, the BrEaST dataset contains lesions with obvious morphological diversity, further contributing to the fluctuations under the same experimental partitioning strategy. However, even under these challenging conditions, our method demonstrates better performance compared to other approaches on the BrEaST dataset. In summary, the proposed challenge-aware strategy demonstrates its adaptability and effectiveness across three diverse datasets: AMUBUS, BUSI, and BrEaST. This consistency across multiple modalities validates the robustness and generalization ability of CAU-net, making it a powerful and reliable approach for breast ultrasound segmentation tasks.

Table 5
Quantitative results (mean ± std) on BUSI. The best results are highlighted in red, while the secondary ones are marked in blue.

Method	Acc (%)↑	Se (%)↑	Dice (%)↑	IoU (%)↑	HD (mm)↓
U-net [16]	97.12 ± 0.45	88.56 ± 1.84	84.61 ± 2.20	75.16 ± 2.15	8.44 ± 0.52
SegNet [29]	96.53 ± 0.43	87.56 ± 2.37	82.37 ± 1.80	73.58 ± 1.27	8.62 ± 0.46
DeepLabV3+ [39]	95.70 ± 1.36	86.46 ± 3.58	81.67 ± 3.06	71.29 ± 2.85	9.63 ± 1.21
Unet++ [40]	97.12 ± 0.44	89.09 ± 1.29	85.70 ± 1.52	76.11 ± 1.24	8.46 ± 0.49
PraNet [41]	96.83 ± 0.93	88.11 ± 2.62	85.68 ± 2.17	76.36 ± 1.97	8.71 ± 0.63
RFNet [42]	96.74 ± 0.68	88.62 ± 1.38	86.52 ± 2.74	78.24 ± 2.06	7.88 ± 0.45
TRUnet [45]	97.13 ± 0.42	89.72 ± 0.96	86.13 ± 1.59	77.61 ± 1.12	8.50 ± 0.51
PSPNet [43]	97.28 ± 0.46	89.22 ± 1.18	85.89 ± 1.96	75.27 ± 2.01	7.81 ± 0.71
nnUnet [44]	97.30 ± 0.31	89.55 ± 1.01	86.10 ± 1.61	77.24 ± 2.03	7.16 ± 0.38
MedNeXt [46]	96.41 ± 0.42	89.33 ± 1.12	86.20 ± 1.72	77.84 ± 2.14	8.37 ± 0.49
ATFE-Net [50]	96.32 ± 0.11	82.78 ± 0.66	82.46 ± 0.37	69.73 ± 0.42	9.68 ± 1.22
AAU-net [47]	96.30 ± 0.23	81.10 ± 0.52	77.51 ± 0.68	63.28 ± 3.03	10.91 ± 1.54
HAU-net [51]	96.80 ± 0.16	80.37 ± 1.34	83.11 ± 2.07	75.26 ± 2.08	10.67 ± 2.44
Hybrid-CAF [52]	94.04 ± 0.41	80.10 ± 0.15	83.42 ± 0.72	72.56 ± 0.85	9.32 ± 0.88
CAU-net	97.48 ± 0.38	89.83 ± 1.29	87.27 ± 1.44	79.13 ± 1.08	7.73 ± 0.44

Table 6
Quantitative results (mean ± std) on BrEaST. The best results are highlighted in red, while the secondary ones are marked in blue.

Method	Acc (%)↑	Se (%)↑	Dice (%)↑	IoU (%)↑	HD (mm)↓
U-net [16]	96.65 ± 1.80	80.16 ± 2.83	74.79 ± 2.65	65.80 ± 2.72	9.16 ± 2.49
SegNet [29]	96.88 ± 1.95	81.37 ± 3.74	74.99 ± 3.22	65.57 ± 3.62	9.38 ± 2.73
DeepLabV3+ [39]	95.13 ± 1.84	79.46 ± 2.94	74.35 ± 3.26	64.27 ± 3.29	10.31 ± 2.48
Unet++ [40]	96.25 ± 1.69	81.50 ± 2.21	75.42 ± 3.73	65.68 ± 3.15	9.94 ± 2.48
PraNet [41]	97.24 ± 1.19	80.85 ± 2.36	76.46 ± 2.85	67.68 ± 2.92	9.12 ± 1.51
RFNet [42]	97.55 ± 1.20	81.68 ± 2.47	76.78 ± 3.62	67.41 ± 3.67	8.87 ± 2.49
TRUnet [45]	97.93 ± 1.37	82.35 ± 2.71	77.23 ± 3.02	67.89 ± 3.51	8.82 ± 2.33
PSPNet [43]	97.45 ± 1.52	81.21 ± 1.56	76.91 ± 3.28	66.83 ± 2.84	9.88 ± 2.12
nnUnet [44]	97.42 ± 0.77	80.69 ± 2.04	76.53 ± 2.15	67.47 ± 2.18	9.15 ± 1.62
MedNeXt [46]	97.76 ± 0.93	82.45 ± 2.38	77.20 ± 2.97	68.03 ± 3.13	8.77 ± 2.65
AAU-net [47]	97.81 ± 0.82	81.02 ± 2.73	76.36 ± 2.85	66.59 ± 3.62	9.38 ± 2.91
EMCAD [48]	97.50 ± 0.94	80.05 ± 2.88	75.69 ± 3.54	65.17 ± 4.12	9.52 ± 3.03
MoSID [49]	96.98 ± 1.17	78.97 ± 3.16	73.85 ± 4.01	63.92 ± 4.55	10.07 ± 2.87
CAU-net	98.28 ± 0.84	82.79 ± 2.60	77.88 ± 3.19	68.55 ± 3.56	8.55 ± 2.83

Table 7
Computational complexity comparisons of different methods.

Methods	Parameters	FLOPs
U-net [16]	17.27M	40.16G
DeepLabV3+ [39]	39.76M	14.92G
Unet++ [40]	9.16M	34.65G
PraNet [41]	32.55M	6.93G
TRUnet [45]	93.19M	128.68G
nnUnet [44]	30.58M	7.42G
MedNeXt [46]	95.74M	26.20G
ATFE-Net [50]	24.96M	6.06G
CAU-net	30.18M	7.33G

In addition, our method has the appropriate computational efficiency, which is suitable for clinical applications while maintaining competitive performance. Table 7 compares the parameters (Params) and floating-point operations (FLOPs) of our method with the classical and the state-of-the-art methods, providing a comprehensive evaluation. Our method has an appropriate scale of parameters, ensuring that it does not bring excessive computational cost. Here, Unet serves as the most basic baseline for this type of research, while Unet++ is specifically designed for fast computation, resulting in fewer parameters but a relatively inferior performance compared to recent methods. On the other hand, TRUnet and MedNeXt have certain segmentation performances, but their networks are very complex with huge size of parameters. Other models have similar parameter amounts and FLOPs to ours, yet our model achieves better results on all performance metrics across the three evaluation datasets. Therefore, in summary, our method not only demonstrates better overall performance but also has good potential for clinical applications.

Finally, in order to further verify the contribution of each branch in the proposed model, the results of each challenge are compared in detail separately during the reasoning process. Considering that some of the earlier methods are not suitable for the more challenging BUS classification tasks and have limited effectiveness, here, we compare only three new methods, as shown in Table 8. The results clearly demonstrate that the proposed CAU-net framework achieves a superior overall segmentation performance in terms of Dice coefficient and HD values for all challenges. In summary, the combination of challenge awareness strategies with adaptive integration of specific branches of different challenges improves segmentation performance. Thus, our CAU-net can effectively handle complex cases with many challenges, demonstrating its robustness and potential for a wider range of ultrasound image segmentation applications.

4.5.2. Qualitative comparison

Fig. 4 showcases the visual segmentation results by different models. The first column displays the input images, followed by segmentation results from different approaches, with our CAU-net results in the final column. Ground truth contours are marked in red, while predicted contours are marked in green. The rows represent distinct segmentation challenges: (a) small tumors (ST), (b) noise interference (NI), and (c) blurred boundaries (BB).

- **Small Tumors (ST):** The top line describes the challenges of small tumors (ST). In this case, the tumor is not only difficult to be located, but also easily confused with the surrounding glands. Unlike other models, our CAU-net successfully avoids false positives when dealing with ST challenges and accurately identifies tumor regions, even in low-contrast areas, while maintaining clear boundaries.
- **Noise Interference (NI):** The second line shows ultrasound images that are obviously disturbed by noise (NI). As shown in

Table 8
Comparison of results from different methods on three challenge subsets. The best results are highlighted in red, while the secondary ones are marked in blue.

	ST		NI		BB	
	Dice (%)	HD (mm)	Dice (%)	HD (mm)	Dice (%)	HD (mm)
TRUnet	81.86	6.52	82.15	7.08	82.34	7.12
nnUnet	83.67	6.27	83.25	6.79	83.42	6.90
MedNeXt	83.55	6.31	84.09	6.98	83.63	6.94
CAU-net	85.27	6.14	84.61	6.85	84.39	6.83

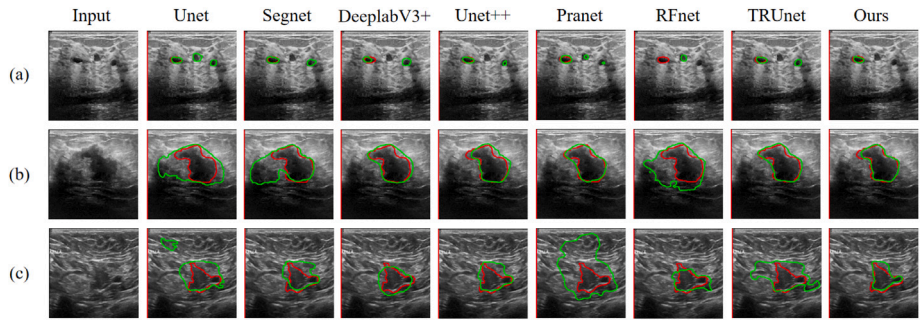

Fig. 4. Visual segmentation results produced by different segmentation methods. The ground truth is plotted in red, and the prediction result is plotted in green. Rows (a) to (c) present the challenge sample of ST, NI, and BB respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 9
The segmentation results (mean) on DDTI. The best results are marked in bold. The best results are highlighted in red, while the secondary ones are marked in blue.

Method	Acc %	Se %	Dice %	IoU %	HD (mm)
U-net [16]	93.40	82.26	74.39	63.28	11.21
SegNet [29]	94.68	86.17	77.54	67.08	10.82
DeepLabV3+ [39]	94.31	86.24	79.51	68.88	11.06
Unet++ [40]	94.43	87.65	80.12	70.47	10.60
PraNet [41]	94.83	87.11	81.48	71.36	10.62
RFNet [42]	94.74	87.62	81.91	71.84	10.58
TRUnet [45]	95.13	87.90	82.45	72.38	10.52
PSPNet [43]	94.38	85.16	78.62	70.81	10.55
nnUnet [44]	94.47	85.28	80.06	71.34	10.41
MedNeXt [46]	94.28	87.51	81.34	72.57	10.25
CAU-net	95.63	87.78	82.63	72.97	10.68

the last figure in the second row, our method does a good job for eliminating interference and achieving accurate segmentation, especially the "neck" of the lesion is more evident in our result.

- **Blurred Boundaries (BB):** The last line is the segmentation result of a typical image with blurred boundaries, demonstrating the ability of CAU-net to accurately separate the lesion, without being interfered by confused adjacent pixels. It can effectively separate the lesion from the fuzzy adjacent tissue.

In summary, as shown in the figures, our method can accurately segment tumors in the presence of the three challenges, which validates the effectiveness of the proposed CAU-net.

4.6. Generalizability verification

To further validate the applicability of our network for ultrasound image segmentation, we also conduct additional experiments on a thyroid ultrasound nodule segmentation dataset (DDTI). DDTI has some similar problems with tumor segmentation in BUS images. The dataset consists of 637 thyroid ultrasound images. We split the dataset into a training set and a testing set with a 4:1 ratio. The experimental setup is the same as the previous experiments. The segmentation results of different methods on DDTI are shown in Table 9. Since ultrasound imaging of different organs is somewhat different, their challenges are also relatively varied, which influences the performance of the network to some extent. However, our method consistently achieves superior performance on most metrics, which validates the ability of CAU-net to segment the lesion in different types of ultrasound images.

5. Discussion

5.1. Failure cases and limitations

The proposed CAU-net is effective for handling challenging samples such as BB, NI, and ST. However, there are a few cases where the network fails. For instance, the network struggles in the case when the lesion is small and similar to the surrounding background, meanwhile, the surrounding gland occupies a larger area. Consequently, the network tends to incorrectly identify the glands, as evidenced in the first row of Fig. 5. In addition, it is not necessary that large lesions can be located well. The limitation of ultrasound imaging results in an uneven distribution of internal intensity when the lesion is large, which leads to holes in the segmented result, as depicted in the second row of Fig. 5. Moreover, as described in Section 3, we employ a two-stage training strategy to enable the loss of training samples to be backpropagated through all three challenging branches, which leads to the network not being trained end-to-end. Further, the computational cost of our method is higher than that of the other compared methods due to the three challenge branches used in the network. Therefore, our future research will aim to address the limitations mentioned above and enhance the robustness of the method.

5.2. Application

Breast cancer is a severe affliction that impacts women's health and safety. Therefore, automated methods for lesion segmentation in BUS images play a crucial role in computer-aided diagnosis (CAD). Our work aims to address the major challenges in ultrasound image segmentation and achieve accurate tumor segmentation. As shown in Section 4, the quantitative results and visual results demonstrate the effectiveness of the proposed CAU-net. Our ongoing collaboration with local hospitals has enabled us to address the problem of segmentation algorithms from a radiologist's diagnostic perspective. However, in clinical practice, it is more common for radiologists to pick out a few

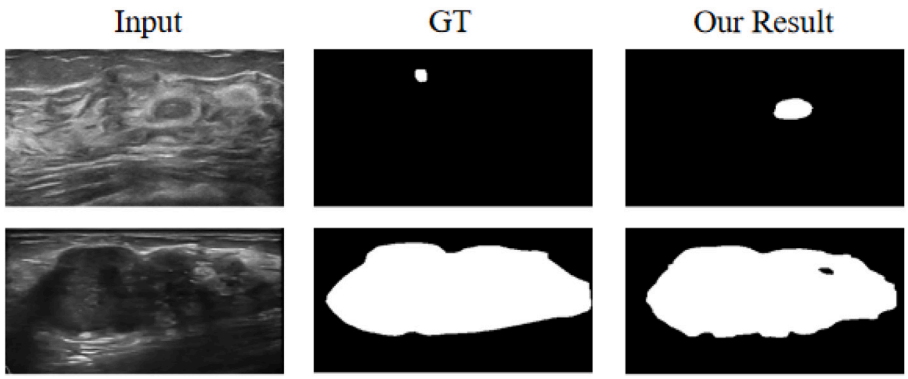

Fig. 5. Failure cases in AMUBUS.

keyframes from an ultrasound video to make a diagnosis. To select the exact keyframe, they sometimes have to look back and forth, which is also a time-consuming part of the diagnosis. It will further improve the efficiency of diagnosis if this process could be automated. Therefore, in the upcoming research, besides improving the shortcomings of the existing work, we will also extend the research on BUS images to BUS videos, realizing the breast lesion segmentation in videos.

6. Conclusion

In this paper, a novel challenge-aware U-net (CAU-net) is proposed for lesion segmentation in BUS images. In the network, we design a challenge-aware backbone to learn the appearance representation of samples from the three challenges. Moreover, an adaptive aggregate layer (AAL) is well-designed to enable the network to handle different samples with arbitrary challenges. To further improve the accuracy of segmentation, a graph reasoning module (GRM) is introduced into the network to learn global information in BUS images. We compare our network with several state-of-the-art methods on two datasets. The results demonstrate that our method achieves better performance than others. Moreover, the visual segmentation results also show the excellent capability of our method in BUS image segmentation. In the future, we will take into account the insufficiency and limitations of existing public BUS datasets, and further improve our dataset and release it. On the other hand, we plan to expand the dataset to more practical video-form data and develop the video segmentation method based on it.

CRediT authorship contribution statement

Dengdi Sun: Methodology. **Changxu Dong:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology. **Yuchen Yan:** Writing – original draft. **Bo Jiang:** Investigation. **Yayang Duan:** Data curation. **Zhengzheng Tu:** Writing – review & editing. **Chaoxue Zhang:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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